

#FABCONSQLCON2026

FABCON

Microsoft Fabric
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ATLANTA MARCH 16 - 20, 2026

Mastering Fabric Data Agents: From Setup to Success



Marc Lelijveld
Wednesday March 18th

Personas

Customer

Financial Advisors

Portfolio Managers

...

Modern Technology

Chat with your Data

Predictions

Data Management
Platform



Fabric

Platforms

Digital Experience

CRM & ERP

Document
Management

...

Let's Define Success



Reliable answers



Secured



Well adopted



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 Data-Marc.com



Exploring solution areas

What are the alternatives to copilot?

What can you conclude from the orders placed versus the orders shipped?

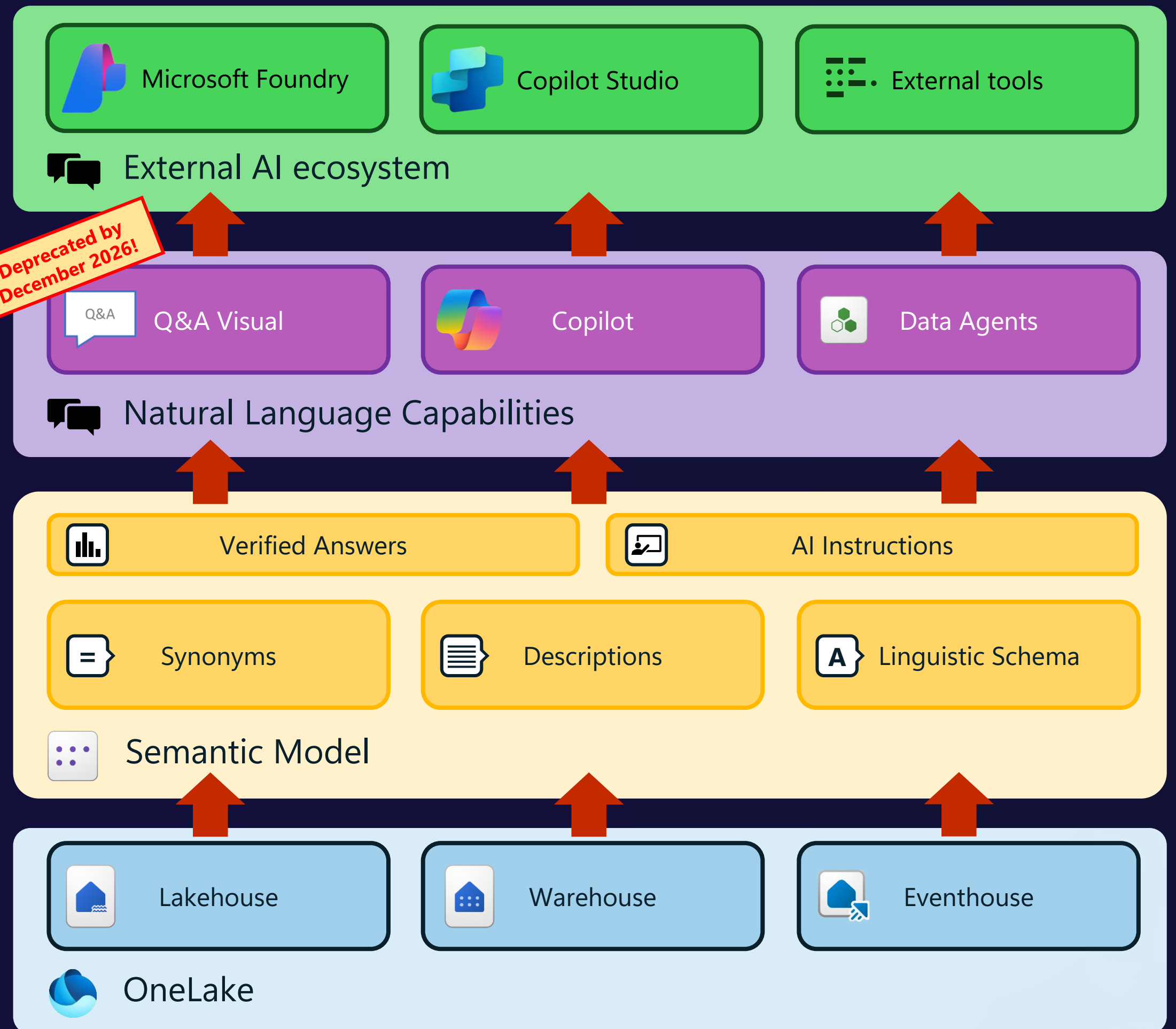


Overview

End-to-end flow



Deprecated by
December 2026!



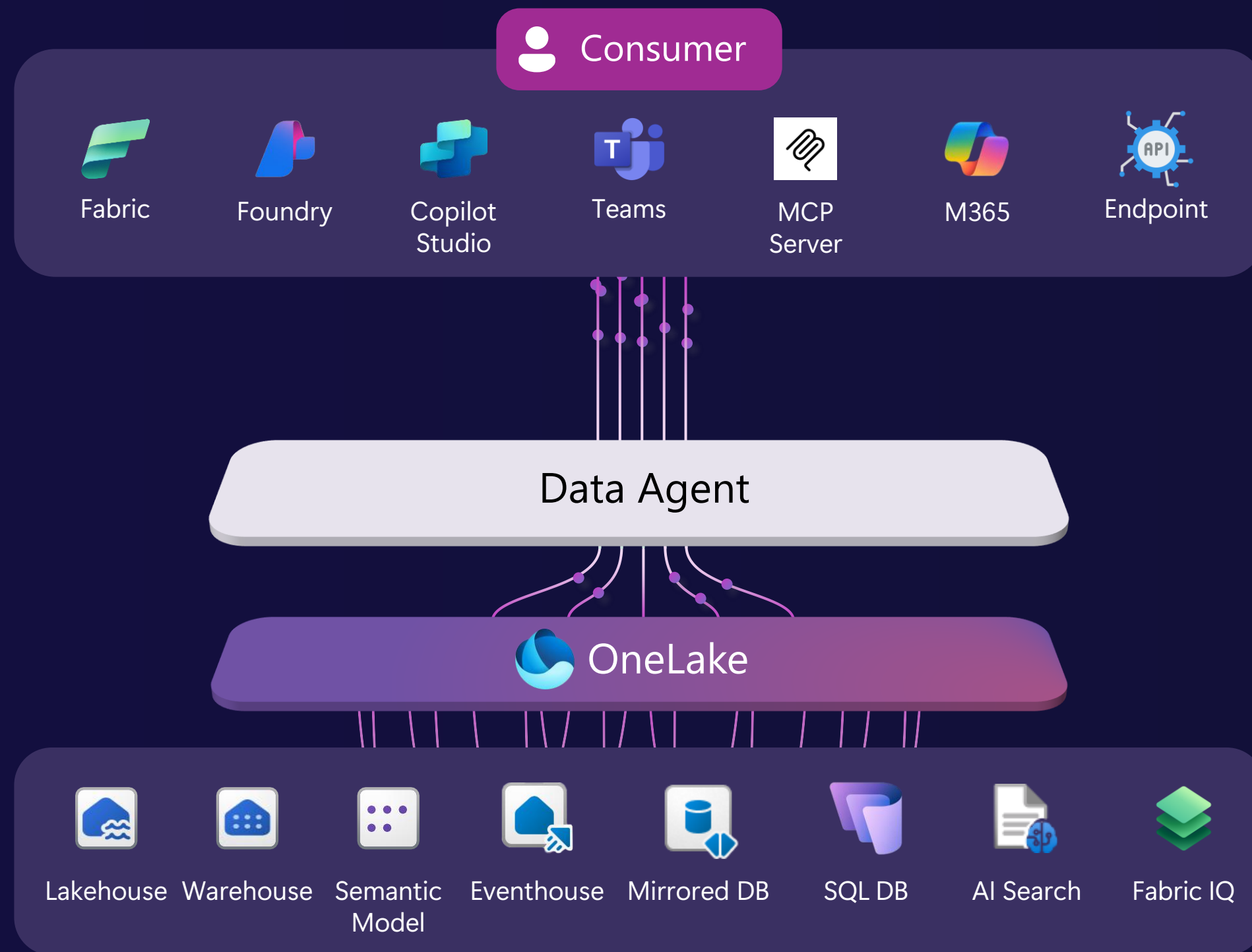


Deep Dive **Fabric Data Agent**

Prerequisites for Data Agents

- Fabric paid capacity (any size)
- Fabric Data Agent tenant setting is enabled
- Cross-geo processing & storing for AI is enabled
- Copilot tenant switch is enabled
- A valid source (Warehouse, Lakehouse, Power BI semantic model or KQL Database)
- Power BI semantic models via XMLA endpoints tenant switch is enabled

Fabric Data Agents



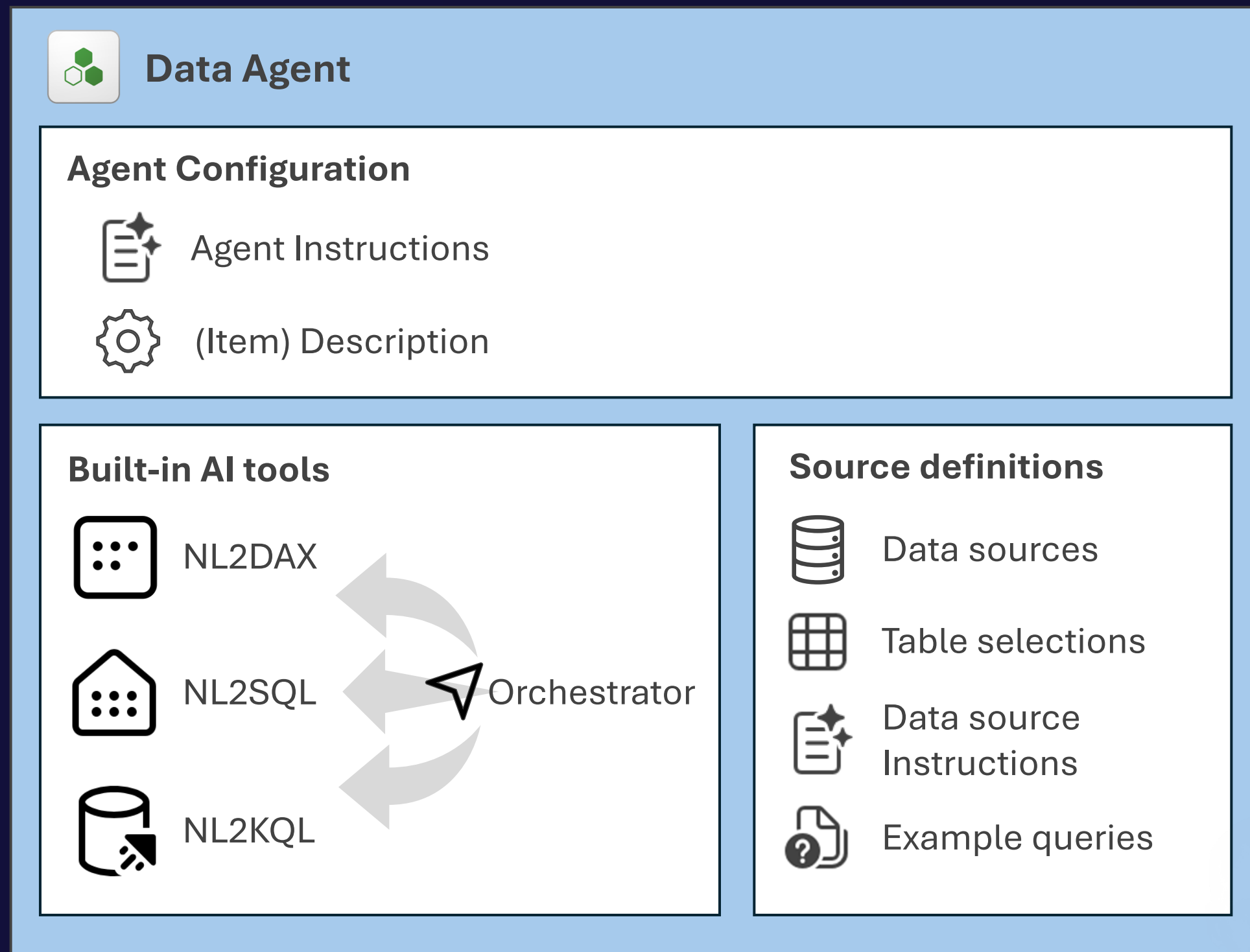
A **virtual analyst** that allows users to interact with and gain insights from enterprise data in OneLake

 **New** Consume your data agents in **M365 Copilot** and as **remote MCP Server** in **VS Code**

 **New** Data agents now support unstructured data through **Azure AI Search**

 **New** Data agents now support **Fabric IQ Workload** through **Ontology**

Data Agent building blocks



Demo

Data Agent Config

The screenshot displays the Microsoft Fabric Data Agent configuration page. The top navigation bar includes the URL 'Data-Marc.com', the 'Fabric' logo, a 'Fuel analysis' tab, and a status indicator 'Allowed to be shared external'. A search bar is located on the right. The left sidebar contains navigation links for Home, Workspaces, Copilot, OneLake catalog, Monitor, and FabCon ATL. The main content area is titled 'Home' and features a toolbar with icons for saving, refreshing, settings, adding data, agent instructions, publishing, and reverting to the published version. The 'Explorer' pane on the left shows a tree view with 'Data' and 'Setup' sections. Under 'Setup', there are links for 'Agent instructions', 'LH_Store_Gold' (expanded), 'Data source instructions', and 'Example queries'. The 'Agent instructions' section is active, displaying three main configuration areas: 'Purpose', 'Accuracy Rules', and 'Response Style'. The 'Purpose' section states the goal is to answer business questions using connected enterprise data sources. The 'Accuracy Rules' section lists three rules: only use information from connected data sources, do not fabricate data, and ask a clarifying question if the request is ambiguous. The 'Response Style' section lists four rules: clear and concise, written in professional business language, focused on the direct answer first, and structured when useful (short summary + optional table). The 'Planning Approach' section lists five steps: understand user intent, select the most appropriate data source, use existing business measures, validate the result, and return the answer with a short explanation if needed. On the right, a 'Test the agent's response' section shows three example queries and their results, each with a '1 step completed' status. The bottom status bar indicates the user is logged in as 'main'.

Home

Workspaces

Copilot

OneLake catalog

Monitor

FabCon ATL

Home

Save Refresh Settings Add data Agent instructions Publish Revert to published version

Explorer

Data Setup

Agent instructions

LH_Store_Gold

Data source instructions

Example queries

Agent instructions

Guide the data agent to the best data sources for different types of questions by writing a detailed system prompt. [Learn more](#)

Purpose

Answer business questions using the connected enterprise data sources. Provide clear, concise, and accurate insights based only on available data.

Accuracy Rules

- Only use information from connected data sources.
- Do not fabricate data.
- Ask a clarifying question if the request is ambiguous.

Response Style

Responses should be:

- Clear and concise
- Written in professional business language
- Focused on the direct answer first
- Structured when useful (short summary + optional table)

Planning Approach

When answering a question:

- Understand the user intent and identify the required metric, dimension, and time period.
- Select the most appropriate data source.
- Use existing business measures where available.
- Validate that the result logically matches the question.
- Return the answer with a short explanation if needed.

Test the agent's response

The current (latest) miles driven by the car.

Converted to miles: 115,352 km ÷ 1.60934

1 step completed

The average price difference between the two cities.

1 step completed

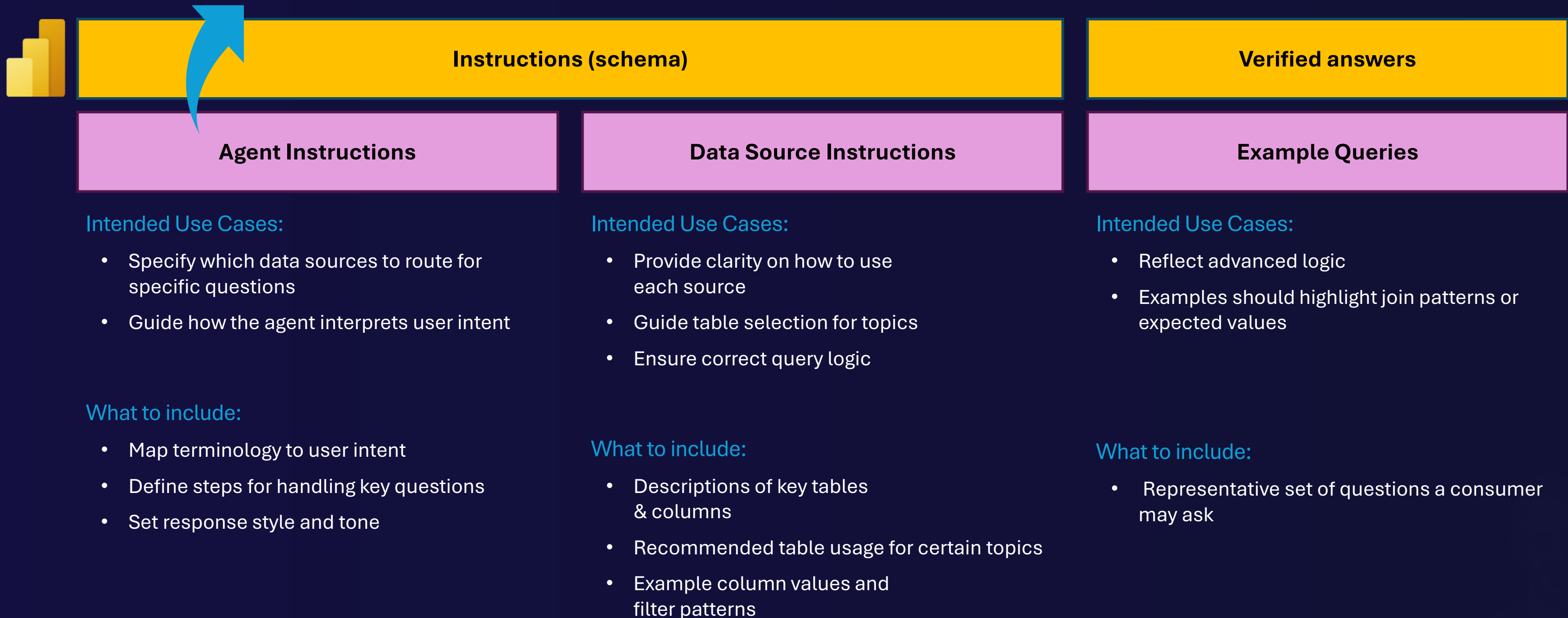
The current (latest) miles driven by the car. Please provide more details, please let me know.

1 step completed

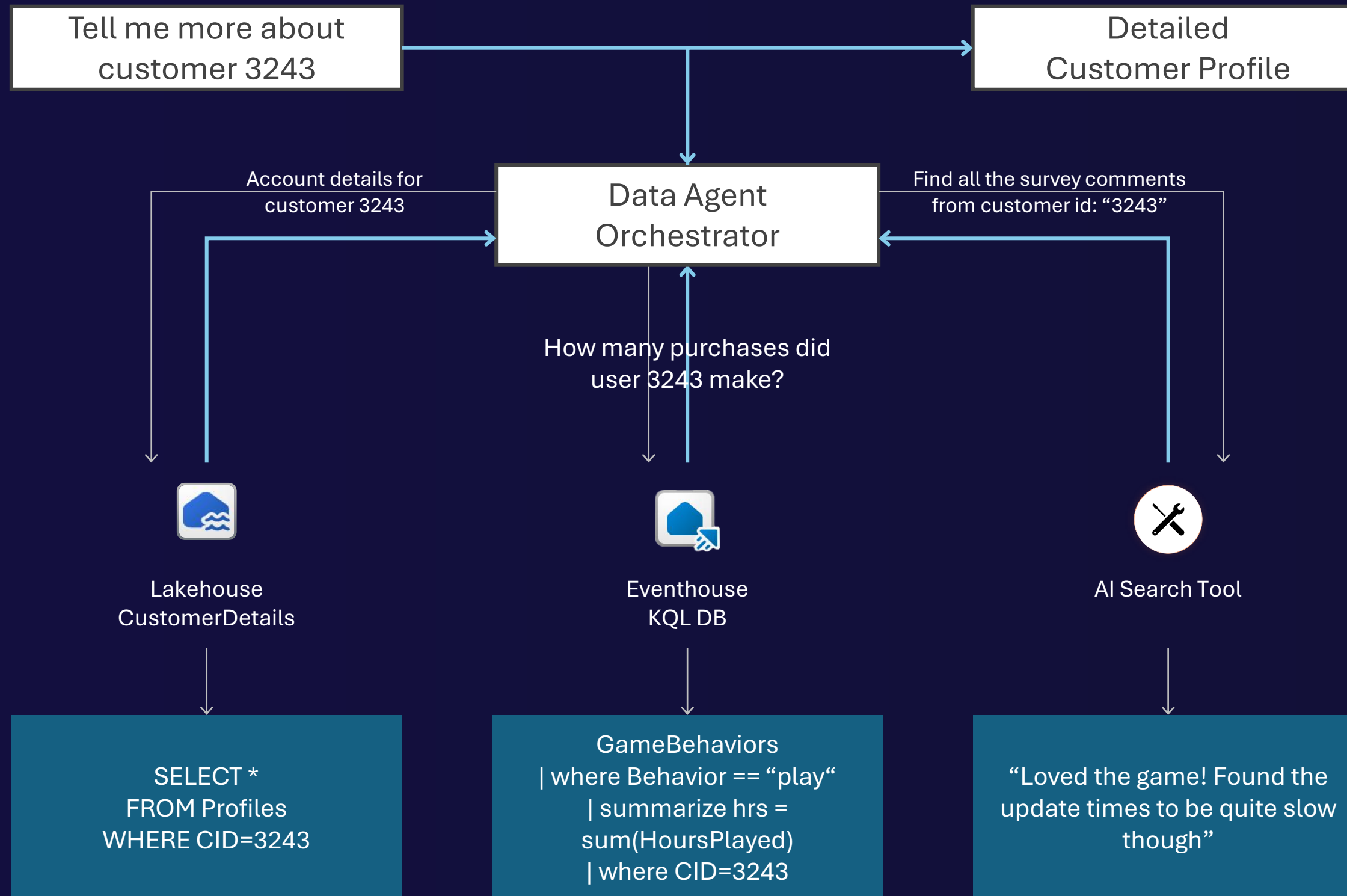
Ask a question to test the agent's response.

Created with AI. Mistakes are possible.

Creator configurations



The Data Agent flow



- 1 The data agent delegates questions to specific tools or data sources.
- 2 Data Sources are specialized at taking a natural language question and generating a response – query or result.
- 3 Data Sources & Tools supply their response back to the orchestrator – allowing it to process the information and think about the next step or return the summarized result.

Reasoning over multiple sources

Decent instructions allows the Data Agent to reason over multiple data sources

Here are the average advised retail prices for Euro 95 (BenzineEuro95) by year, including the year-to-date value for 2026:

- 2006: €1.37
- 2007: €1.41
- 2008: €1.48
- 2009: €1.35
- 2010: €1.50
- 2011: €1.64
- 2012: €1.76
- 2013: €1.74
- 2014: €1.69
- 2015: €1.56
- 2016: €1.48
- 2017: €1.55
- 2018: €1.62
- 2019: €1.65
- 2020: €1.56
- 2021: €1.81
- 2022: €2.07
- 2023: €1.92
- 2024: €1.97
- 2025: €1.92
- 2026 (year-to-date): €2.04

This provides a reliable year-by-year view, including the current average for 2026 up to today.

👍 👎

1 step completed ^

⌚ Response time: 10 sec

✓ Analyzed LH_Store_Gold Lakehouse for: "What is the average advised retail price of Euro 95 (BenzineEuro95) for each year, including a year-to-date average for the current year 2026?"

How does that relate to the paid prices for the years we have fuel transactions for? And only focus on Euro 95.

Here is how the average paid price for Euro 95 at the pump (based on actual fuel transactions) compares with the advised retail price for years where transaction data is available:

Average paid price for Euro 95 by year:

- 2020: €1.58
- 2021: €1.89
- 2022: €2.14
- 2023: €1.91
- 2024: €1.98
- 2025: €1.88
- 2026 (year-to-date): €1.99

Key observation:

- The paid price is typically slightly lower than the advised retail price each year, reflecting common discounts and promotions at the pump.
- For 2026 (year-to-date), the paid price (€1.99) remains just below the advised retail price (€2.04).

If you'd like a direct side-by-side comparison in table form, let me know!

👍 👎

1 step completed ^

⌚ Response time: 26 sec

✓ Analyzed Fuel Analysis SemanticModel for: "What is the average paid price for Euro 95 per year, for all years where there were fuel transactions for Euro 95? Only show Euro 95. Use year-to-date for th..."

Combined reasoning

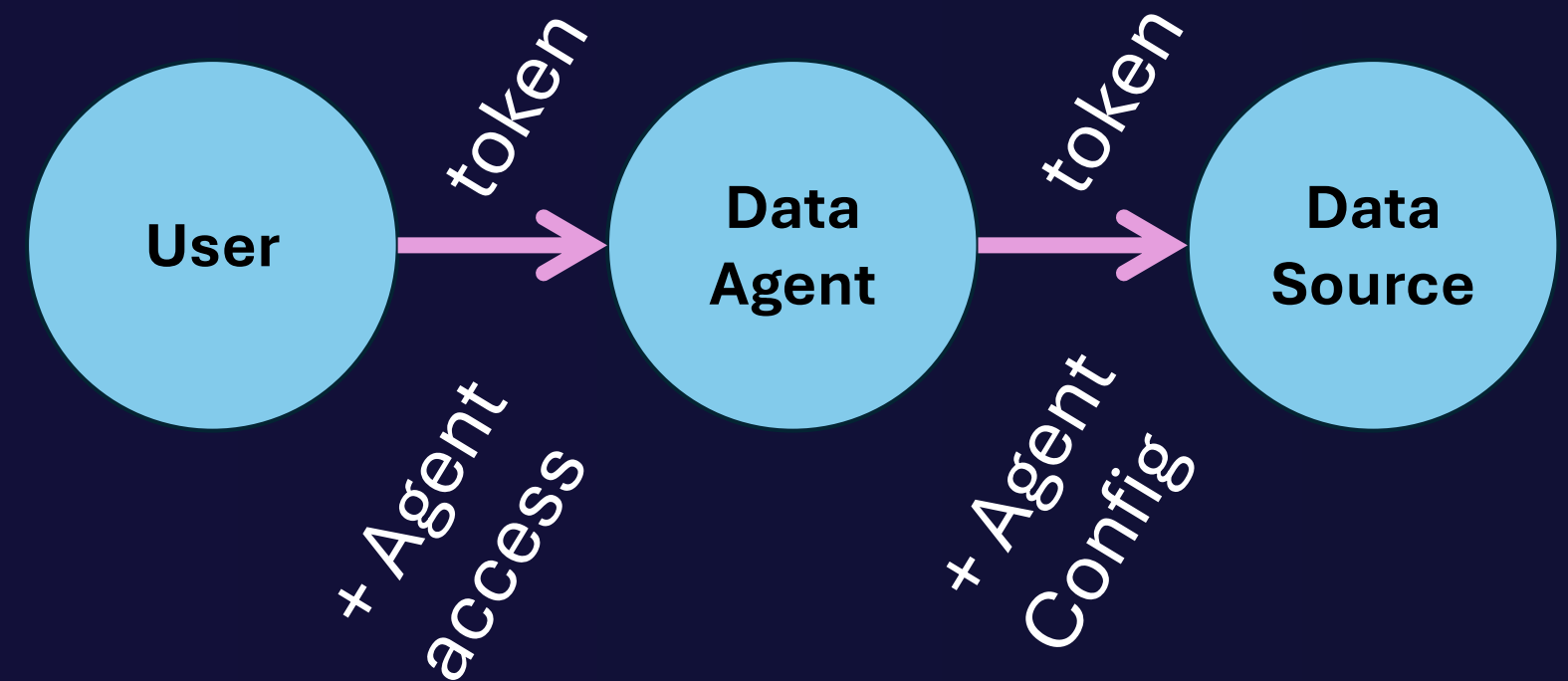
Semantic Model

Security & Permissions – Data Sources

Pass-through mechanics

- User credentials flow to the data source(s)
- Permission evaluated per-user, per-query
- RLS/OLS etc. are baked-in and are inherited from the source(s)

Service principals (SPN) are **NOT** supported, Access requires a user identity.



Security & Permissions – Data Sources

Source Type	Minimum Permission	Reasoning
Semantic Model	Read (new!, previously build)	Agent generates DAX queries
Lakehouse	Read on the lakehouse item (and table access if enforced)	Write not required unless modifying data.
Warehouse	Read (SELECT on relevant tables)	Higher permissions only for DML/DDDL operations.
KQL database	Reader role on the database	Higher roles only for management commands.
Ontology	Read on the ontology item, and Read on the underlying semantic model, lakehouse, or KQL database bound to the ontology	
Other supported sources	Query/read-level access	Must allow metadata + data retrieval.

Different types of supported sources



Lakehouse



Warehouse



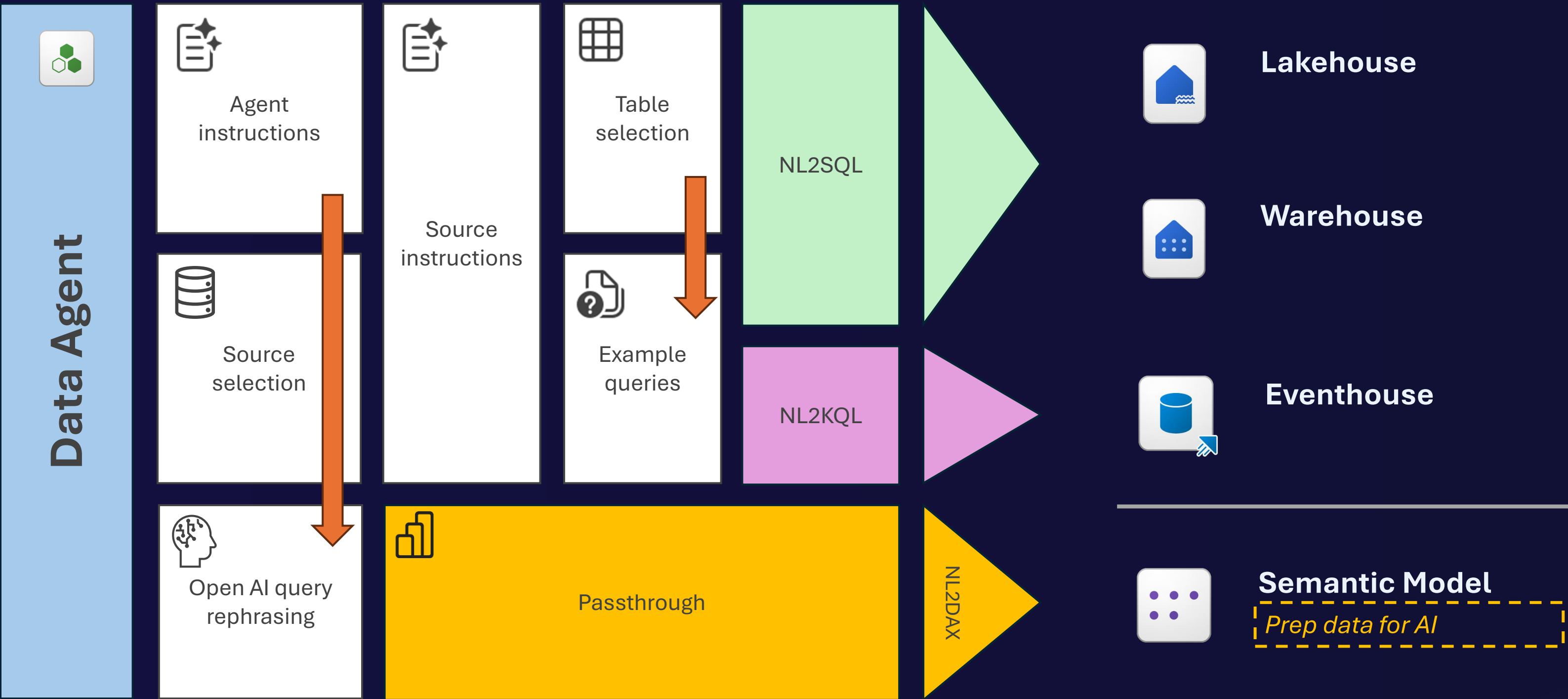
Eventhouse



Semantic Model

And related sources (like mirrored sources)

Data Agent Orchestrator - Inner workings





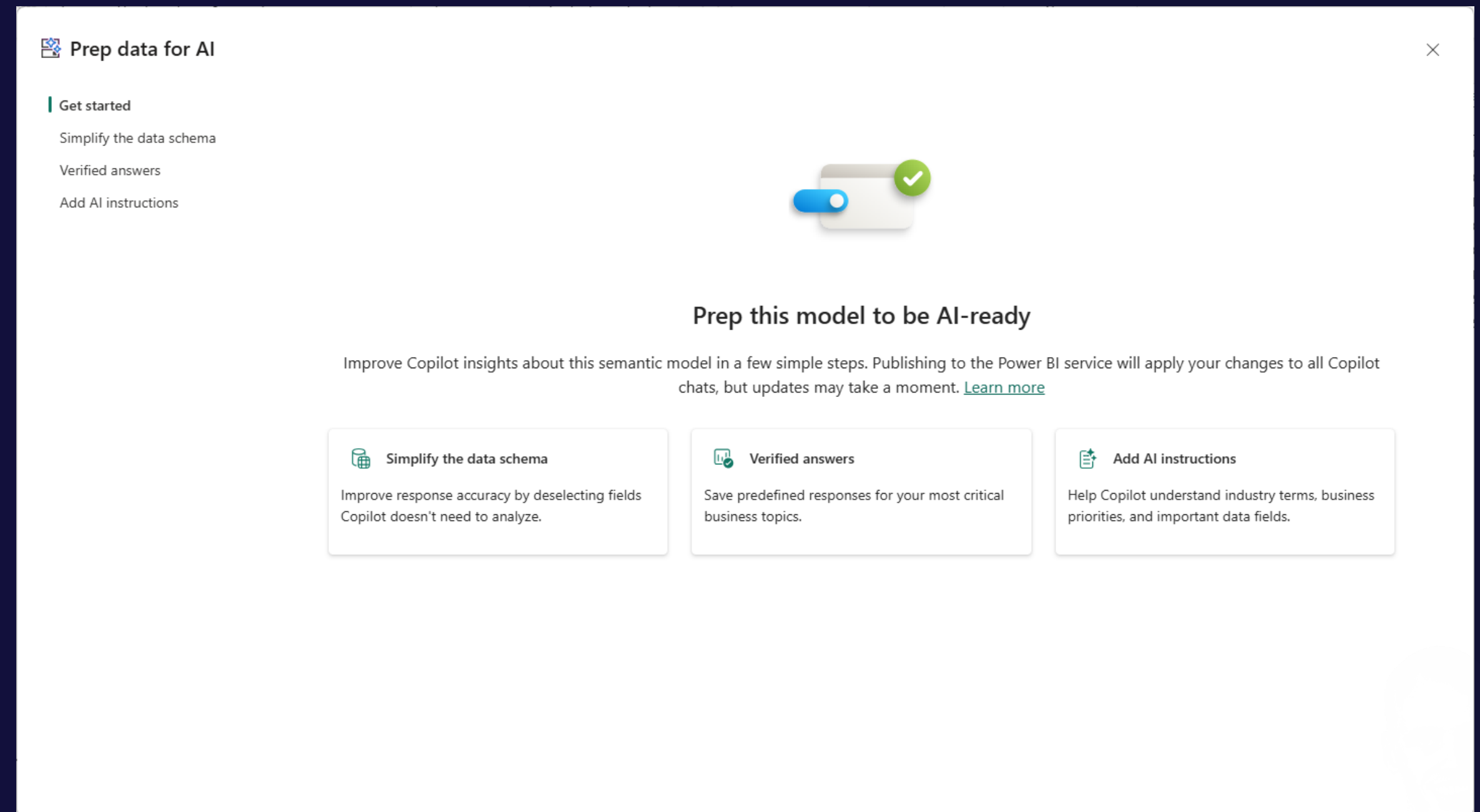
Semantic Models & Prep Data for AI



Semantic models: Prep data for AI

Three main pillars

- Simplify schema
- Verified answers (links to visuals)
- AI Instructions

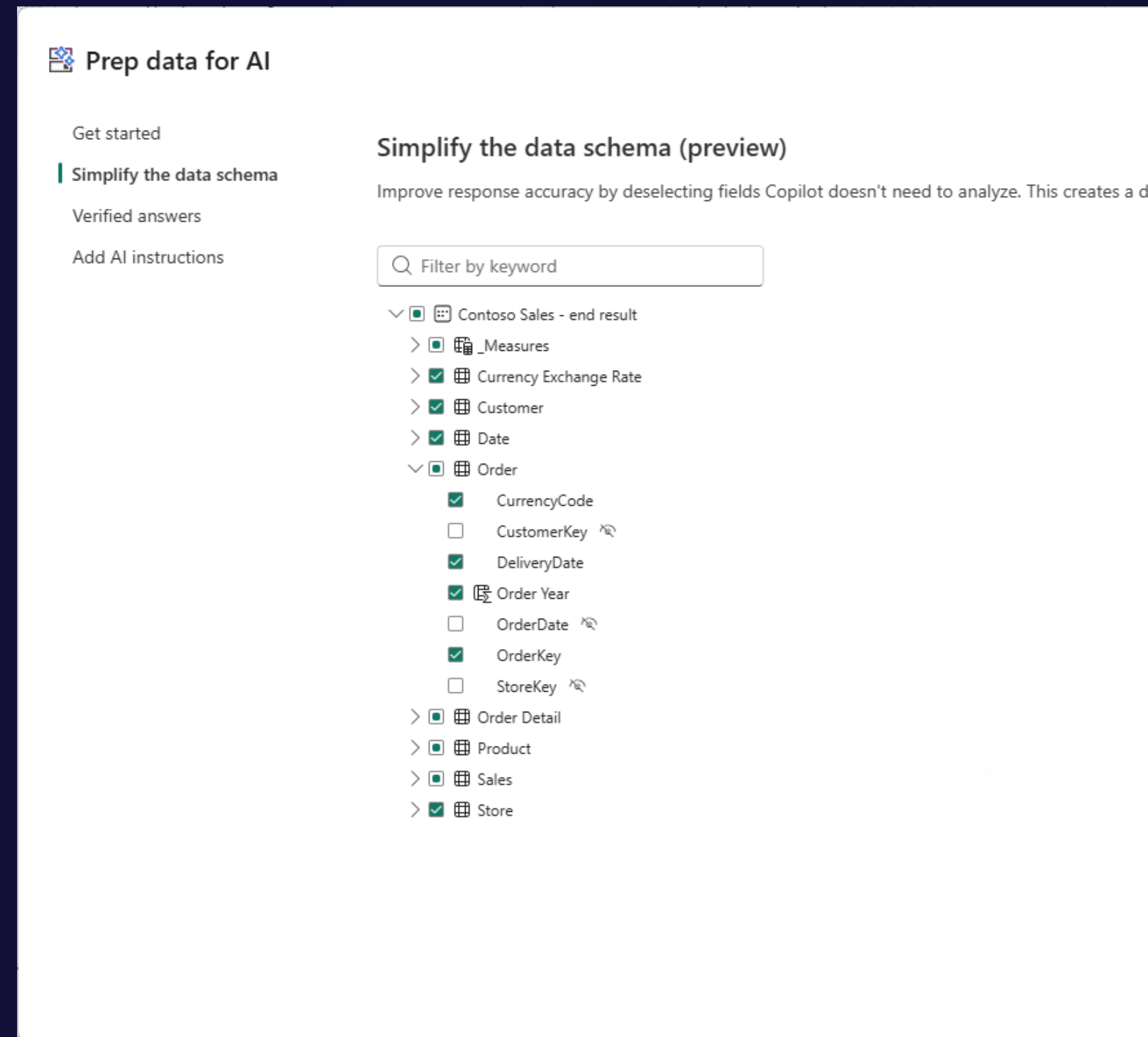




Semantic models: Prep data for AI – Simplify schema

- Only keep what is relevant!
- Hide report-specific measures
 - Conditional formatting
 - Measures outputting texts like visual titles
- Disconnected tables likely cause issues
→ build relationships!

Keep in mind that Visual Calculations are not part of the semantic model!



Semantic models: Prep data for AI – Verified answers

Prep data for AI

Get started

Simplify the data schema

Verified answers

Add AI instructions

Verified answers (preview)

← All verified answers

Phrases connected to verified answers

Add phrases people might use when asking about the model data, and Copilot will respond with this visual. Test your phrases out in the chat before publishing.

What are my top 3 performing product categories

Add

⊕ Show the top 3 categories by sales.

Copilot suggestions

Which product categories have the highest sales?

What are sales by category?

↻ Refresh

Copilot uses AI. Always review content for mistakes. [Read terms](#)

Visual ⓘ

The visual below will appear in Copilot's responses to phrases you add. Any changes to the report won't impact this visual.

Sales Amount

0.5bn

0.4bn

0.3bn

0.2bn

0.1bn

0.0bn

Computers

Cell phones

Home Appliances

CategoryName

Apply

Close

FABCON

SQLCON

ATLANTA26

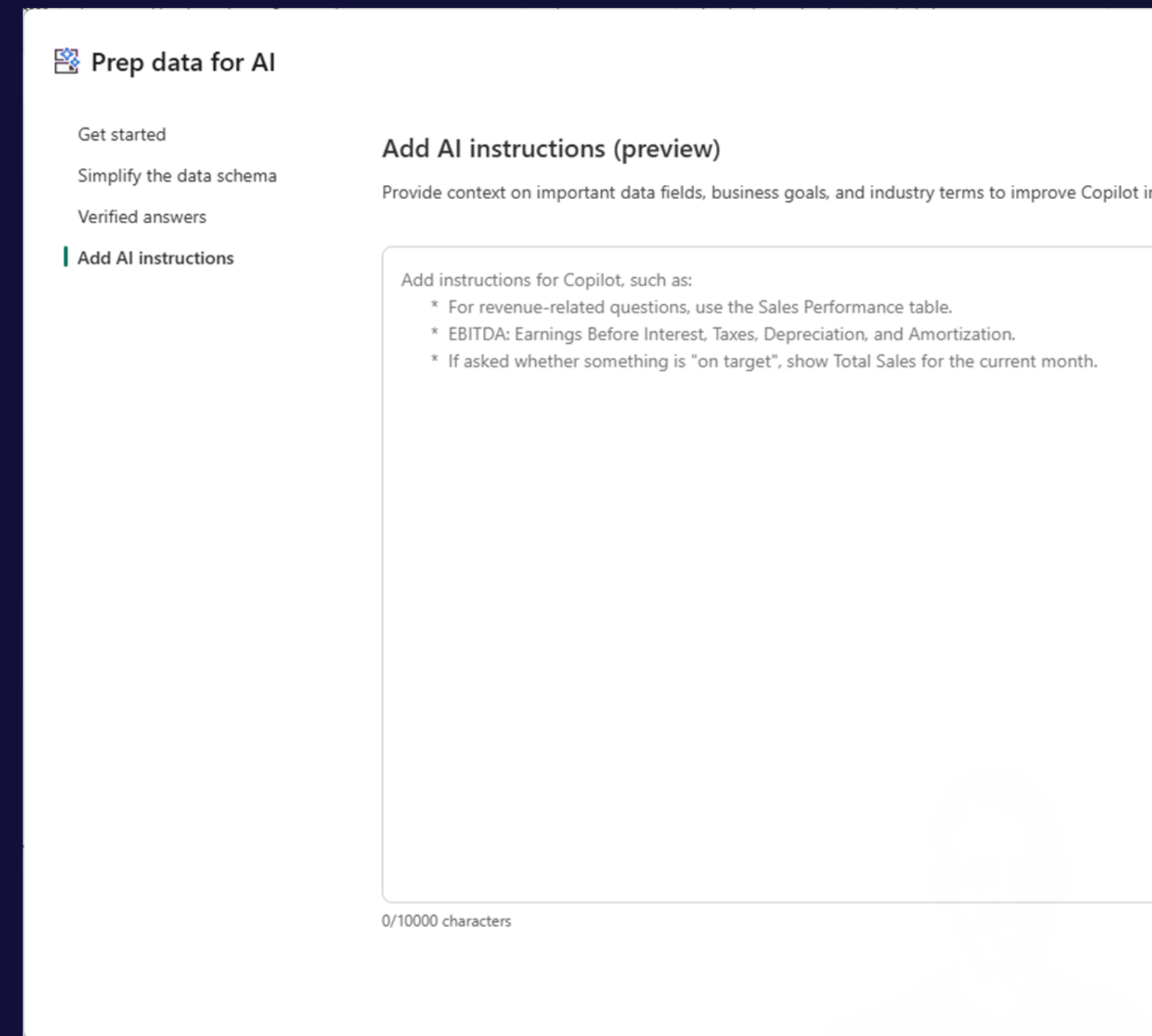
JOIN THE
CONVERSATION

#FABCONSQLCON26



Semantic models: Prep data for AI – AI instructions

- Unstructured guidance (textual)
- No guarantee the guidance is followed exactly – see it as a hint!
- Response formatting
 - Style
 - Tone
- Business specific logic – for example:
 - Define what is a top performer – high or low?
 - Peak / off-season: which months are these?





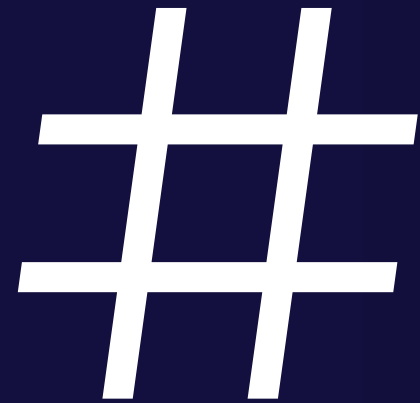
Common pitfalls

- Conflicting or overly **complex** instructions
- Skipping **schema refinement** – especially with large semantic models containing many tables
- Not using a **star schema** (resulting in inefficient DAX)
- Relying on **hidden fields**
- Including **unnecessary measures**: Simplify the schema,
 - Remove measures that are not relevant or broken,
 - Hide report measure (for conditional formatting for example),
 - Avoid duplicating logic in different measures
- **Non-descriptive naming** (technical names for objects)
- Relying on **implicit measures**: high chance of inconsistency!
- **Ambiguous date fields** (e.g. Order Date, Ship Date, ...)

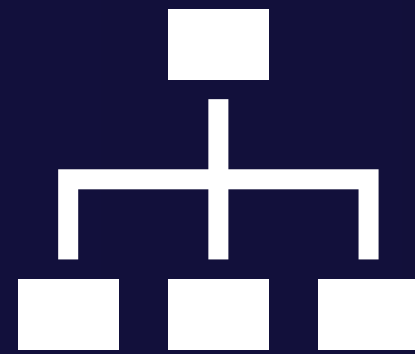




Advantages of using a Semantic Model as a data source



**Allows metadata
enrichment**



**Business logic
(measures)**



**User friendly and
terminology**



**Single source of
truth (incl reports)**



Data Agent Diagnostics

Keep in mind!

The answers will only be as good as your data!

siso

Shit In, Shit Out: If you're putting shit data into an experiment, you're going to get shit data out.

Your results are bubkes, what did you expect? SIS0!



Image from Urban Dictionary: <https://x.com/urbandictionary/status/958044898212630528>

Diagnostics

“AI is wrong” to structured root cause analysis!

Where to start your investigation when the answer is not what you’re looking for?

Permissions

- RLS / CLS
- Item level access
- governance policies

Prompt & grounding

- Clear question
- Proper metadata

Source grounding

- Semantic model, lakehouse, warehouse, eventhouse
- Relationships and item naming

Notes for model

- Tone & voice
- Context (what is a good performer, high or low?)
- Hints (example queries)

Generated query analysis

- Diagnostics

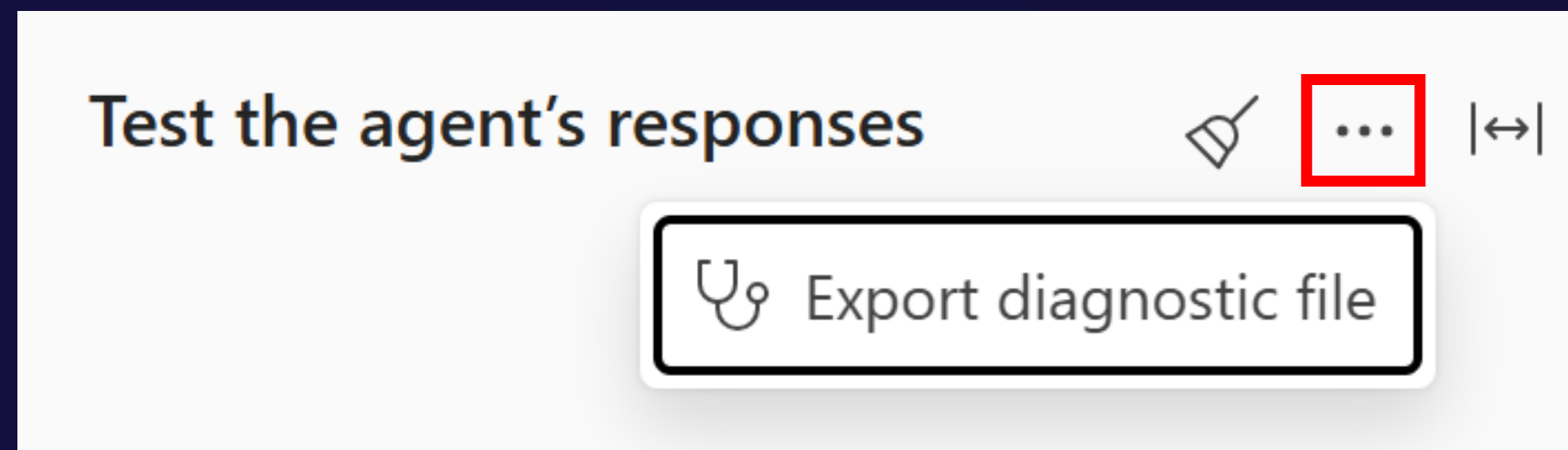
Capacity & performance

- Storage mode (for semantic models)
- Model size
- Load on the capacity

Diagnostics

“AI is wrong” to structured root cause analysis!

Where to start your investigation when the answer is not what you’re looking for?



User prompts

Data source schemas
and configurations

File metadata and
contents

Environment and
artifact identifiers
(GUIDS)

Any proprietary
information you may
have entered

Diagnostics

Source
instructions

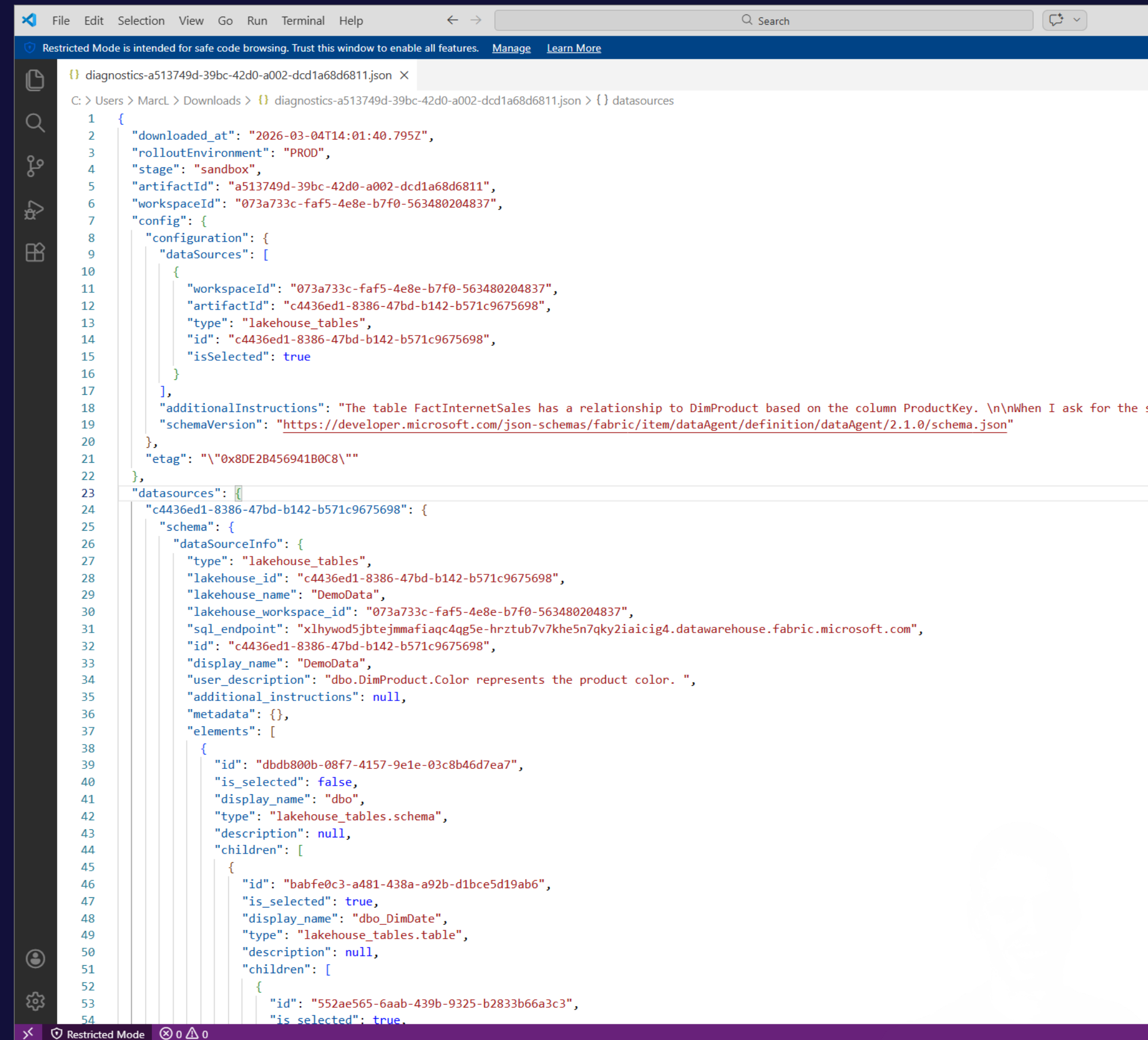
Schema
information

...and much more!

```
{
  "downloaded_at": "2026-03-04T14:01:40.795Z",
  "rolloutEnvironment": "PROD",
  "stage": "sandbox",
  "artifactId": "a513749d-39bc-42d0-a002-dcd1a68d6811",
  "workspaceId": "073a733c-faf5-4e8e-b7f0-563480204837",
  "config": {
    "configuration": {
      "datasources": [
        {
          "workspaceId": "073a733c-faf5-4e8e-b7f0-563480204837",
          "artifactId": "c4436ed1-8386-47bd-b142-b571c9675698",
          "type": "lakehouse_tables",
          "id": "c4436ed1-8386-47bd-b142-b571c9675698",
          "isSelected": true
        }
      ]
    },
    "additionalInstructions": "The table FactInternetSales has a relationship to DimProduct based on the column ProductKey. \n\nWhen I ask for the sales amount, or total sales, or revenue",
    "schemaVersion": "https://developer.microsoft.com/json-schemas/fabric/item/dataAgent/definition/dataAgent/2.1.0/schema.json"
  },
  "etag": "\"0x8DE2B456941B0C8\""
},
"datasources": [
  {
    "c4436ed1-8386-47bd-b142-b571c9675698": {
      "schema": {
        "dataSourceInfo": {
          "type": "lakehouse_tables",
          "lakehouse_id": "c4436ed1-8386-47bd-b142-b571c9675698",
          "lakehouse_name": "DemoData",
          "lakehouse_workspace_id": "073a733c-faf5-4e8e-b7f0-563480204837",
          "sql_endpoint": "xlhywod5jbtejmmafiaqc4qg5e-hrztub7v7khe5n7qky2iaicig4.datawarehouse.fabric.microsoft.com",
          "id": "c4436ed1-8386-47bd-b142-b571c9675698",
          "display_name": "DemoData",
          "user_description": "dbo.DimProduct.Color represents the product color. ",
          "additional_instructions": null,
          "metadata": {},
          "elements": [
            {
              "id": "dbdb800b-08f7-4157-9e1e-03c8b46d7ea7",
              "is_selected": false,
              "display_name": "dbo",
              "type": "lakehouse_tables.schema",
              "description": null,
              "children": [
                {
                  "id": "babfe0c3-a481-438a-a92b-d1bce5d19ab6",
                  "is_selected": true,
                  "display_name": "dbo_DimDate",
                  "type": "lakehouse_tables.table",
                  "description": null,
                  "children": [
                    {
                      "id": "552ae565-6aab-439b-9325-b2833b66a3c3",
                      "is_selected": true
                    }
                  ]
                }
              ]
            }
          ]
        }
      }
    }
  ]
}
```


Demo

Data Agent Diagnostics



```
{
  "diagnostics-a513749d-39bc-42d0-a002-dcd1a68d6811.json": {
    "downloaded_at": "2026-03-04T14:01:40.795Z",
    "rolloutEnvironment": "PROD",
    "stage": "sandbox",
    "artifactId": "a513749d-39bc-42d0-a002-dcd1a68d6811",
    "workspaceId": "073a733c-faf5-4e8e-b7f0-563480204837",
    "config": {
      "configuration": {
        "dataSources": [
          {
            "workspaceId": "073a733c-faf5-4e8e-b7f0-563480204837",
            "artifactId": "c4436ed1-8386-47bd-b142-b571c9675698",
            "type": "lakehouse_tables",
            "id": "c4436ed1-8386-47bd-b142-b571c9675698",
            "isSelected": true
          }
        ],
        "additionalInstructions": "The table FactInternetSales has a relationship to DimProduct based on the column ProductKey. \n\nWhen I ask for the s",
        "schemaVersion": "https://developer.microsoft.com/json-schemas/fabric/item/dataAgent/definition/dataAgent/2.1.0/schema.json"
      },
      "etags": "\0x8DE2B456941B0C8\"
    },
    "datasources": [
      {
        "c4436ed1-8386-47bd-b142-b571c9675698": {
          "schema": {
            "dataSourceInfo": {
              "type": "lakehouse_tables",
              "lakehouse_id": "c4436ed1-8386-47bd-b142-b571c9675698",
              "lakehouse_name": "DemoData",
              "lakehouse_workspace_id": "073a733c-faf5-4e8e-b7f0-563480204837",
              "sql_endpoint": "xlhywod5jbtejmmafiac4qg5e-hrztub7v7khe5n7qky2iaicig4.datawarehouse.fabric.microsoft.com",
              "id": "c4436ed1-8386-47bd-b142-b571c9675698",
              "display_name": "DemoData",
              "user_description": "dbo.DimProduct.Color represents the product color. ",
              "additional_instructions": null,
              "metadata": {},
              "elements": [
                {
                  "id": "dbdb800b-08f7-4157-9e1e-03c8b46d7ea7",
                  "is_selected": false,
                  "display_name": "dbo",
                  "type": "lakehouse_tables.schema",
                  "description": null,
                  "children": [
                    {
                      "id": "babfe0c3-a481-438a-a92b-d1bce5d19ab6",
                      "is_selected": true,
                      "display_name": "dbo_DimDate",
                      "type": "lakehouse_tables.table",
                      "description": null,
                      "children": [
                        {
                          "id": "552ae565-6aab-439b-9325-b2833b66a3c3",
                          "is_selected": true,

```



Using the Data Agent **Software Development Kit**

Data Agent SDK

- **Programmatic Management:** Create, update, and delete Data Agent artifacts seamlessly from a Fabric Notebook
- **Data Source Integration:** Easily connect to and integrate multiple data sources, for enhanced data analysis and insight generation.
- **OpenAI Assistants API Support:** Use the OpenAI Assistants API for rapid prototyping and experimentation.
- **Workflow Automation:** Automate routine tasks to reduce manual efforts and improve operational efficiency.
- **Resource Optimization:** Optimize configuration and management of Data Agent resources to better align with specific customer needs.

 Copy

```
%pip install fabric-data-agent-sdk
```

Data Agent testing & validation

Gains confidence in the data agent, produces accurate, explainable and repeatable answers

- Establishes trust by validating your Data Agent against a known, authoritative dataset
 - Results are compared one-to-one
 - Enables objective evaluation
-

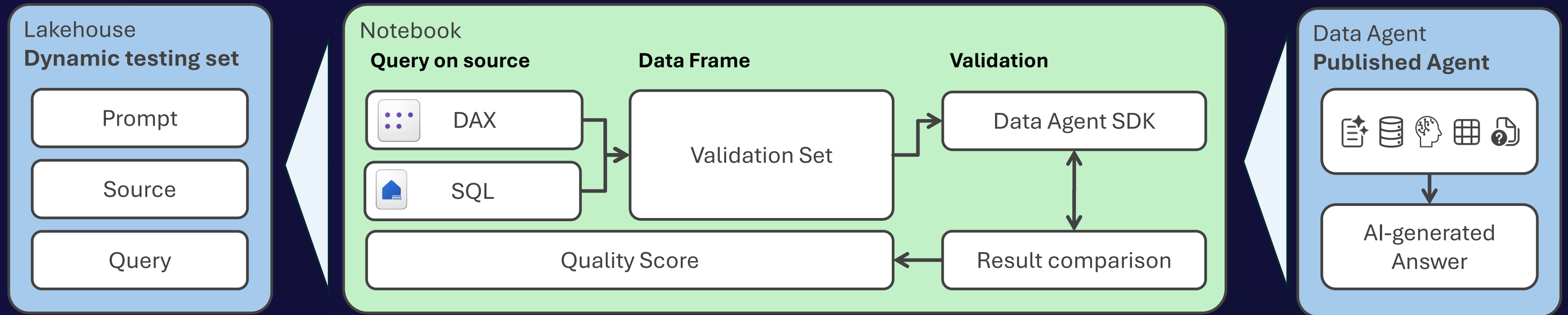
Static / fixed testing

- Predefined values stored in the notebook (dataframe) for testing
- Easy to setup and guarantees results across runs
- Limited flexibility: cannot cover new or changing data
- Mainly used for unit tests or small-scale validations

Dynamic testing

- Stores the query instead of hardcoded values
- Pulls data from the source at runtime for realistic testing
- Enables validation against changing data
- Supports end-to-end checks
- Scales easily with multiple tables or environments, aligning with automated QA pipelines

Dynamic testing



Specifics used:

- **Fabric Semantic Link** for executing DAX on a semantic model
- **Data Agents SDK** connecting and validating prompts in a Data Agent

Demo

Dynamic testing

Home

Workspaces

Copilot

OneLake catalog

Monitor

FabCon ATL

...

Home

Edit

AI tools

Run

View

Run all

Connect

PySpark (Python)

Environment

Workspace default

Data Wrangler

Explorer

Data items

Resources

Connections

Add data items

OneLake

LH_MONIT_DataAgentQuality

Tables

dbo

dynamic_test_output

dynamic_test_output_steps

Files

```
1 # Two tables will be created:
2 # - "<table_name>": contains summary results (e.g., accuracy)
3 # - "<table_name>_steps": contains detailed reasoning and step-by-step execution
4 table_name = "dynamic_test_output"
5
6 # Specify the Data Agent stage: "production" (default) or "sandbox"
7 data_agent_stage = "production"
8
9 # Run the evaluation and get the evaluation ID
10 evaluation_id = evaluate_data_agent(
11     df_dynamic_testset,
12     data_agent_name,
13     workspace_name=workspace_name,
14     table_name=table_name,
15     data_agent_stage=data_agent_stage
16 )
17
18 print(f"Unique ID for the current evaluation run: {evaluation_id}")
```

[7] ✓ - Command executed in 48 sec 130 ms by Demo User on 3/12/2026, 6:20:47 PM

The result does not match the provided ground truth(s):

- [thread_y852g5rxFjCxHKKxinyTgeSl](#)
- [thread_TNTx2WQC3YHb0vlp3w9xbFzj](#)

100%|██████████| 4/4 [00:31<00:00, 7.78s/it]
Unique ID for the current evaluation run: e8b2e0b9-4a99-4437-8236-e13314f560f6

Get test results

```
1 # Retrieve a summary of the evaluation results
2 df_summary = get_evaluation_summary(table_name=table_name, verbose=False)
3 print(df_summary)
```

[8] ✓ - Command executed in 9 sec 502 ms by Demo User on 3/12/2026, 6:20:57 PM

index	evaluation_id	T	F	?	%
0	30d1086f-e217-444d-9d95-d0deff3f50f2	1	1	1	33.3
1	e8b2e0b9-4a99-4437-8236-e13314f560f6	2	1	1	50.0

Fabric

Not connected

AutoSave: On

Copilot completions: Off

Result validation & scoring

Evaluation judgement outcomes

- True: Result matched expectations
- False: Result did not meet expectations
- Unclear: Likely the data source not selected/available

index	question	expected_answer	actual_answer	evaluation_judgement
0	What is the current mileage on R-306-ZJ?	115352	The current mileage on R-306-ZJ is 115,352 kilometers.	True
1	What is the price difference between retail and paid price?	0.0191	The difference between the average retail price and the average paid price for fuel in the most recent complete period (February 2026) is 0.0742 per unit. This means, on average, the paid price was 0.0742 lower than the retail price.	False
2	What is the price increase for Euro 95 between 2010 and now?	0.7410000000000001	No data was found for the average retail price of Euro 95 in 2010 or for the most recent period. This may be due to data not being available for those years or due to restrictive filters. If you have a specific timeframe or additional details, please provide them, or let me know if you would like to try a different approach.	Unclear

Fuel Analysis

car

Brand

Car Key

Cars

Cars active

End Date

License Plate

Start Date

Type

date

transactions

LH_Store_Gold

dbo

car

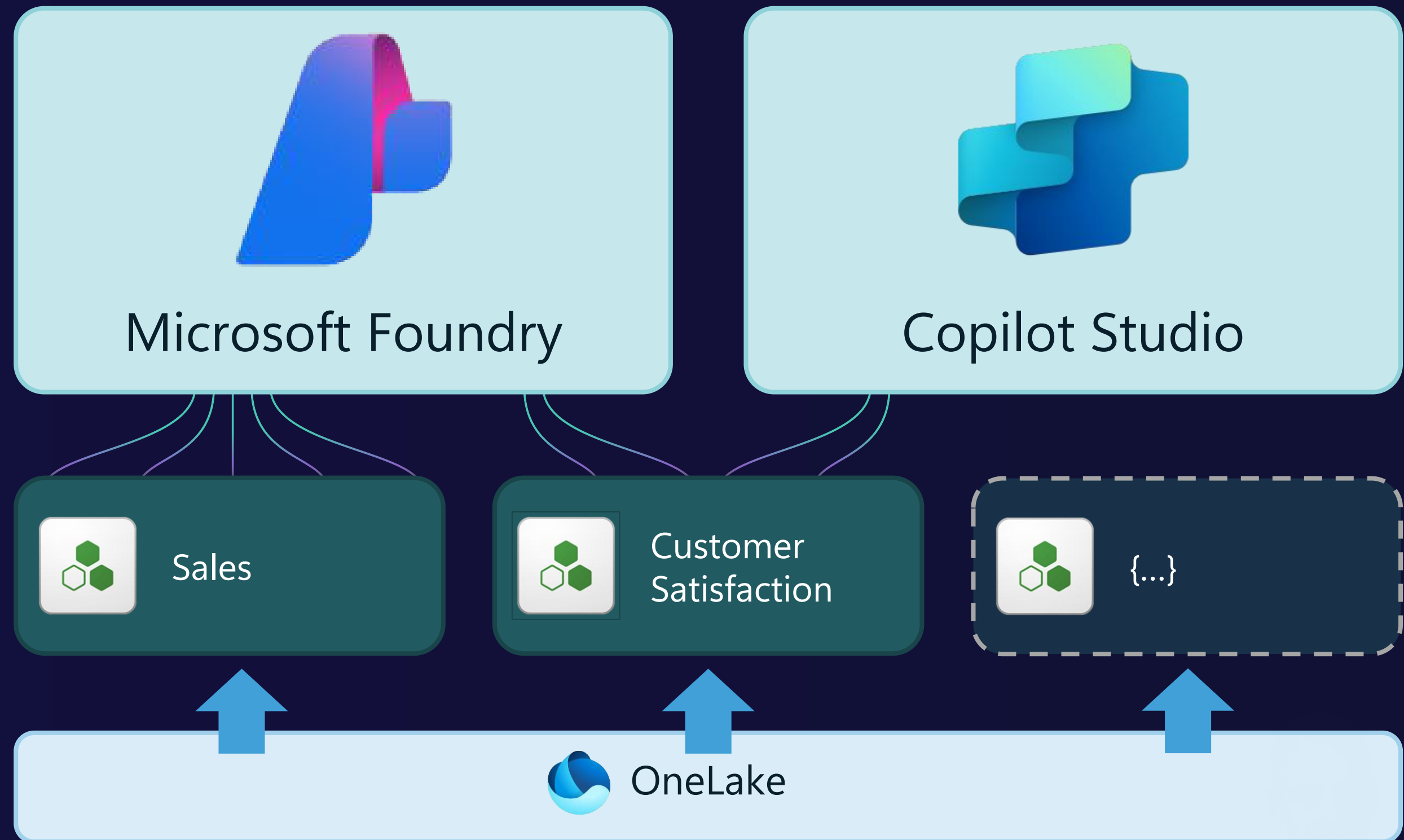
date

recommended retail price

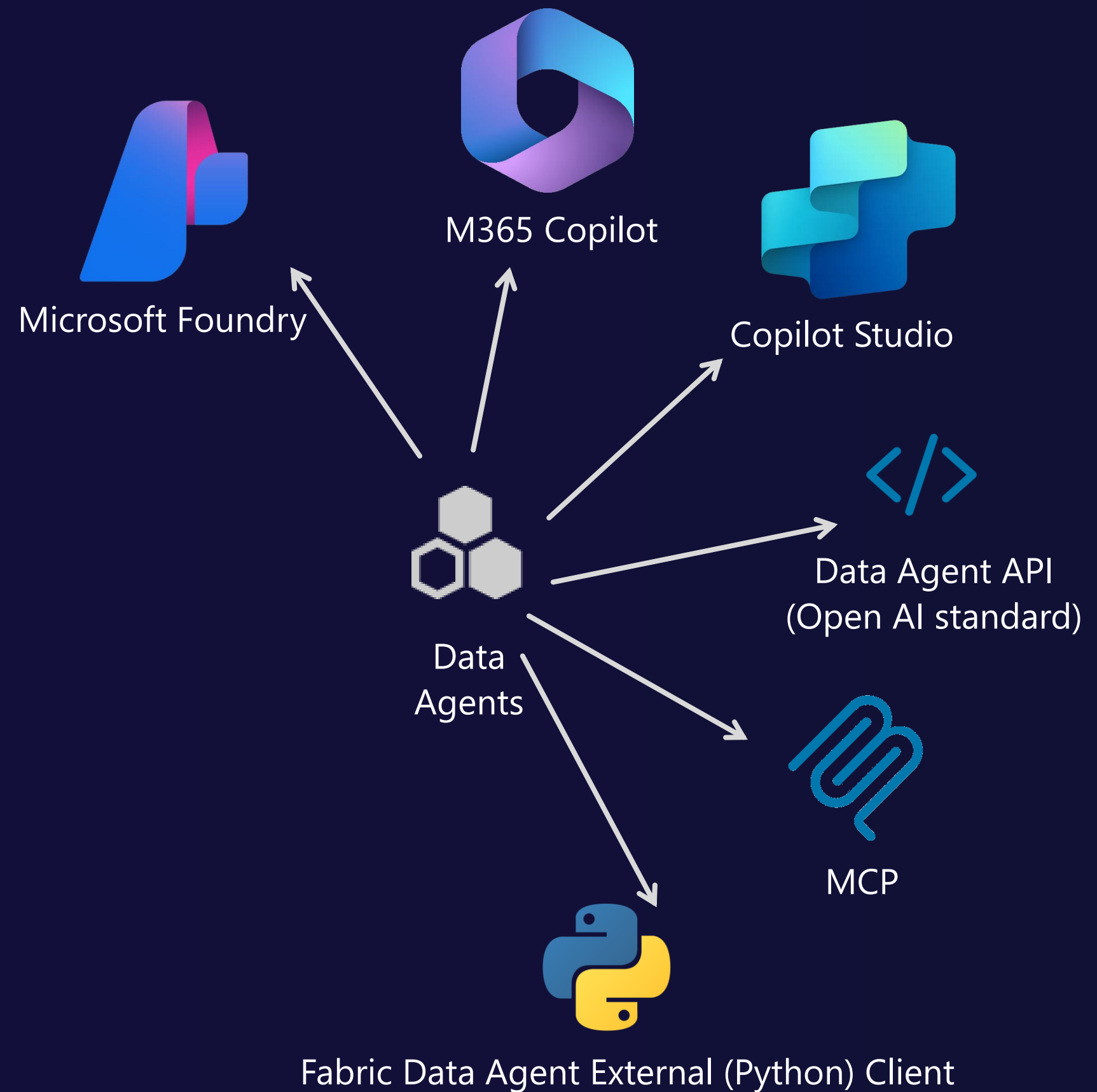
transactions

Agent Architecture

One to rule them all?



Part of a larger
Ecosystem



Recipe to successful implementation



Reliable answers

Through decent
preparation &
(automated) testing



Secured

Inheritance and
passthrough of
permissions



Well adopted

Measured and tested
with diagnostics

Wrap-up

- Fabric Data Agents unlock **governed access to enterprise data**, enabling users to ask questions in plain language across OneLake, semantic models, lakehouses, warehouses, and eventhouses.
- **Strong foundations matter most**, with well-prepared schemas, verified answers, and clear AI instructions driving accuracy and trust.
- **Security and governance are built-in**, with pass-through permissions, RLS/OLS inheritance, and per-user evaluation ensuring answers are both secure and compliant.
- Diagnostics, testing, and validation turn “AI is wrong” into insight, enabling structured root-cause analysis, quality scoring, and continuous **improvement through SDK-driven automation**.
- Data Agents fit into **a broader ecosystem**, integrating with Fabric, Power BI, M365, Copilot Studio, Microsoft Foundry, and custom applications.

Key takeaway:

Reliable AI answers come from reliable data, thoughtful design, and continuous validation.

Relevant Resources

Data Agent tenant settings

<https://learn.microsoft.com/en-us/fabric/data-science/data-agent-tenant-settings>

Data Agent sharing

<https://learn.microsoft.com/en-us/fabric/data-science/data-agent-sharing>

Data Agent SDK

<https://learn.microsoft.com/en-us/fabric/data-science/fabric-data-agent-sdk>

Data Agent Evaluation (testing)

<https://learn.microsoft.com/en-us/fabric/data-science/evaluate-data-agent>

Data Agent Best Practices

<https://learn.microsoft.com/en-us/fabric/data-science/data-agent-configuration-best-practices>

Semantic Model Best Practices for Data Agents

<https://learn.microsoft.com/en-us/fabric/data-science/semantic-model-best-practices>

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